



HEAT AND THERMODYNAMICS — FULL MIND MAP

ऊष्मा और ऊष्मागतिकी · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

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4. Humidity, Air-Conditioning & Evaporative Cooling
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1. Thermodynamic Laws, Heat Transfer & Latent Heat

- Ecosystem me trophic level badhne par usable energy decline ko second law of thermodynamics explain karta hai—energy transformations me useful energy ka kuch hissa dissipate hota hai. [Q1]
- Solid, liquid aur gas tino phases ka ek saath equilibrium me coexist karna triple point kehlata hai; water ka triple point 273.16 K (0.01°C) hota hai. [Q2]
- Radiation ko medium ki zarurat nahi hoti; isi liye Sun se Earth tak heat vacuum ke through aa sakti hai. Convection sirf fluids—liquids aur gases—me bulk motion se hota hai. [Q3–Q4]
- Thermos flask heat exchange ko minimum karta hai: vacuum conduction/convection ko suppress karta hai aur reflective/silvered walls radiation ko reduce karti hain. [Q5, Q57]
- 100°C steam same-temperature boiling water/air se zyada severe burn de sakti hai because condense hote waqt latent heat of vaporization release karti hai (~540 cal/g exam value). [Q6–Q8]
- Steam at 100°C aur water at 100°C me heat content same nahi; latent heat of fusion of ice ~80 cal/g aur vaporization ~540 cal/g. Air-conditioner room se evaporator par heat leta hai aur condenser par bahar reject karta hai. [Q9]
- 1 g ice at 0°C → water at 0°C (80 cal) → water at 100°C (100 cal) → steam at 100°C (540 cal): total ≈ 720 cal. [Q10]



EXAM TRAP — Q5 & Q57 — Thermos Flask

- ▶ Vacuum conduction aur convection ko strongly reduce karta hai, lekin electromagnetic radiation vacuum me travel kar sakti hai. Radiation loss silvered/reflective surfaces se reduce hota hai.
- ▶ Isliye “vacuum prevents radiation” ko literal physics statement mat maanna; source answer ko thermos ke combined insulation design ke context me padho.

2. Temperature Scales, Thermometry & Absolute Zero

- Celsius–Fahrenheit relation: $F = (9/5)C + 32$; inverse $C = (F - 32) \times 5/9$. Linear relation hone se averages par bhi $F = (9/5)C + 32$ apply hota hai. [Q11, Q22]
- Celsius aur Fahrenheit readings –40° par exactly equal hoti hain. [Q12–Q14]
- Agar Fahrenheit reading Celsius ki double ho, $F = 2C$ set karne par $C = 160^\circ\text{C}$ aur $F = 320^\circ\text{F}$ milta hai. [Q15]
- Normal human body temperature exam values me $\sim 37^\circ\text{C} \approx 98.4\text{--}98.6^\circ\text{F} \approx 310\text{ K}$; $98.4^\circ\text{F} \approx 36.89^\circ\text{C}$ (~36.9°C). [Q16–Q19]
- Kelvin thermodynamic scale exam level par negative values nahi leta; absolute zero 0 K = –273.15°C theoretically lowest temperature hai. [Q20, Q27]
- Quick conversions: $200^\circ\text{F} \approx 93.3^\circ\text{C}$; $40^\circ\text{C} = 104^\circ\text{F}$; $50^\circ\text{C} = 122^\circ\text{F}$. [Q21, Q23–Q24]
- Cold countries me alcohol thermometric liquid useful hai because uska freezing point mercury se much lower ho sakta hai. Automobile antifreeze ke liye ethylene glycol widely used hai. [Q25–Q26]

⚠ EXAM TRAP — Q27 — Absolute Zero: Exam Model vs Quantum Physics

- ▶ Competitive-exam level par 0 K ko “minimum possible temperature / thermal motion minimum” likha jata hai.
- ▶ Strict quantum physics me particles ki zero-point motion completely vanish nahi hoti; source ka “completely at rest” classical simplification hai.

3. Thermal Expansion, Thermostat & Refrigeration

- Most materials heating par expand karte hain; mass same aur volume increase hone se density generally decrease hoti hai. [Q28]
- Ice melt hone par source answer ke according volume decreases—liquid water, ice se denser hota hai. [Q29]
- Given heating graph me curved liquid Q ne 100°C straight-line P se faster attain kiya. [Q30]
- Thermostat ka role temperature ko automatically regulate/stable maintain karna hai; refrigerator thermostat bhi preset internal temperature maintain karta hai. [Q31–Q35]
- Food refrigeration ke liye source ka safest option ~4°C hai. Frost-free fridge ki low-humidity environment soggy biscuits se moisture remove karke unhe crisp bana sakti hai. [Q36–Q37]
- Drink me ice melt karne ke liye latent heat surrounding drink se absorb karti hai, isliye drink cool hota hai. [Q38]
- Large refrigeration plants ammonia use kar sakte hain; source domestic refrigerators ko CFCs se contrast karta hai. Ammonia ambient temperature par liquefy karne ke liye sufficiently high pressure chahiye, low pressure nahi. [Q39]
- Freezer me excess ice layer poor conductor/insulator ki tarah heat transfer ko hinder karke cooling efficiency ko adversely affect kar sakti hai. [Q40]
- Isotropic solid heating par dimensions sab directions me expand karte hain: glass rod ki length ke saath width bhi increase hoti hai; heated ring ka inner hole bhi expand hota hai. [Q41–Q42]
- Bimetal strip me higher-expansion metal outer/convex side par aata hai. Copper glass se zyada thermally conductive hone ke karan same temperature par touch me hotter feel ho sakta hai; winter me iron wood se colder feel hota hai because hand se heat faster conduct karta hai. [Q43–Q45]

⚠ EXAM TRAP — Q39 — Refrigerant Wording Is Historical

- ▶ Source domestic refrigerators ko CFC refrigerants se contrast karta hai. Ye older/historical framing hai; modern refrigeration me environmental rules ke kaaran multiple non-CFC refrigerants use hote hain.
- ▶ Core exam concept: large industrial systems me ammonia common ho sakta hai; liquefaction/condensation pressure–temperature conditions par depend karti hai.

4. Humidity, Air-Conditioning & Evaporative Cooling

- Warmer air ki water-vapour carrying capacity generally increase hoti hai. Same moisture content par temperature badhne se relative humidity usually decrease hoti hai. [Q46–Q47]
- Source strong-wind night me dew na banne ko fast evaporation se explain karta hai; practical meteorology me wind surface cooling aur near-surface saturation ko bhi disrupt karti hai. [Q48]
- Air-conditioner temperature ke saath humidity ko control/dehumidify kar sakta hai; ordinary air-cooler humidity ko precise maintain nahi karta aur often moisture add karta hai. [Q49–Q50]
- Desert cooler evaporative cooling par based hai: water evaporation latent heat absorb karke air ko cool karti hai. [Q51]
- Refrigerator/AC cooling refrigerant ke pressure-change, expansion aur especially evaporation/phase-change cycle se hoti hai; source option ise compressed gas ke release/expansion ke roop me frames karta hai. [Q52]
- Wet clothes fastest dry honge jab temperature high aur relative humidity low ho; source options me 60°C + 20% RH earliest drying deta hai. [Q53]

⚠ EXAM TRAP — Q48 & Q52 — Simplified Source Explanations

- ▶ Dew strong wind me suppress ho sakti hai mainly because wind radiatively cooled surface ke paas air ko mix karti hai aur saturation/dew-point condition develop nahi hone deti; “evaporation fast” alone incomplete explanation hai.
- ▶ Refrigeration cycle me cooling ka core step low-pressure refrigerant ka evaporation/heat absorption hai; “release of compressed gas” source-option shorthand hai.

5. Boiling/Freezing Point, Clothing & Perspiration

- Common salt add karne se water ka boiling point increase aur freezing point decrease hota hai—boiling-point elevation + freezing-point depression. [Q54–Q55]
- High altitude par atmospheric pressure low hone se water lower temperature par boil karta hai; source question me salt use ko boiling point thoda raise karke cooking aid se link kiya gaya hai, sirf outside temperature low hone se nahi. [Q56]
- White/light clothes incoming radiant energy ka larger fraction reflect aur less absorb karti hain, isliye summer me preferred; winter me dark clothes greater absorption ke liye useful hote hain. [Q58–Q61]
- Perspiration-related water loss atmospheric temperature aur humidity dono se affected hai. Sweat evaporation body se latent heat le kar thermoregulation me major role play karti hai. [Q62–Q63]
- Fan mainly sweat evaporation ko accelerate karke comfort deta hai; earthen pitcher ke pores se water evaporate hota hai aur latent heat absorb hone se inner water cool rehta hai. [Q64–Q65]

6. Newton’s Cooling, Weather & Atmospheric Pressure

- Newton’s law of cooling ke according heat-loss rate surrounding se temperature difference par depend karti hai. Given Churu ($95-48=47^{\circ}\text{C}$) aur Shimla ($71-24=47^{\circ}\text{C}$) cases me equal ΔT ki wajah se source same cooling time choose karta hai. [Q66]
- Body $90\rightarrow 75^{\circ}\text{C}$ in 25 s cool hota hai; next $75\rightarrow 60^{\circ}\text{C}$ drop generally 25 s se zyada lega because temperature difference surroundings se kam hota jata hai aur cooling rate slow hoti hai. [Q67]
- Source heat-wave criterion question me threshold plains ke liye at least 40°C aur hilly regions ke liye at least 30°C diya gaya hai; question ke dono reversed statements incorrect hain. [Q68]
- Summer me “humid heat” high humidity se associated hai, while xeric condition low moisture/low humidity ko indicate karti hai. [Q69–Q70]
- Atmospheric pressure body par large force lagata hai, lekin internal body-fluid/tissue pressures aur external pressure largely balance karte hain; source ise blood pressure slightly more than atmosphere ke wording me deta hai. [Q71]
- Solute add karne se water potential increase nahi, decrease hota hai; evaporation badhne par air me absolute humidity increase ho sakti hai. [Q72]
- Altitude increase $\rightarrow$ atmospheric pressure decrease $\rightarrow$ water ka boiling point decrease; isliye high altitude par water 100°C se below boil kar sakta hai. [Q73–Q74]

⚠ EXAM TRAP — Q71 — Why Atmospheric Pressure Is Not Felt

- ▶ Body ke andar fluids/gases aur tissues external atmospheric pressure ko largely balance karte hain; human body rigid evacuated shell nahi hai.
- ▶ Source ka “blood exerts slightly more pressure” wording oversimplified hai—blood pressure normally atmospheric pressure ke relative (gauge pressure) me quote kiya jata hai.

7. Pressure Cooker — Pressure, Boiling Point & Safety

- Pressure cooker me steam trapped hone se pressure increase hota hai, water ka boiling point $\sim 100^{\circ}\text{C}$ se upar chala jata hai aur food higher cooking temperature par faster cook hota hai. [Q75–Q80]
- Pressure-cooker handle ebonite jaise poor conductor/insulator se bana hota hai taaki heat hand tak na aaye; “ebonite hard hai” true ho sakta hai, but handle-use ka correct explanation insulation hai. [Q81]
- Cooker ki regulated pressure/maximum cooking temperature mainly vent-hole area aur regulator/weight par depend karti hai; once pressure set ho, extra flame primarily steam-generation/escape badhati hai, set temperature ko indefinitely nahi. [Q82–Q83]

8. Water Anomaly, Clouds & Specific Heat

- Water freezing par expand karta hai; tightly corked glass bottle me ice ka larger volume walls par pressure dal kar bottle tod sakta hai. [Q84–Q85]
- Floating ice melt hone par water level same rehta hai, because floating ice apne weight ke equal water displace karta hai aur melted water exactly wahi displaced volume occupy karta hai. [Q86]
- Cloudy nights clear nights se warmer ho sakti hain because clouds outgoing terrestrial infrared radiation ko absorb/re-emit karke net heat loss reduce karte hain. [Q87–Q88]
- Cloud type matters: low thick clouds incoming solar radiation strongly reflect karke cooling kar sakte hain; high thin clouds outgoing IR trap/re-emit karke warming effect de sakte hain. Source ke reversed statements dono incorrect hain. [Q89]
- Water ki specific heat land se much higher hai, isliye oceans temperature slowly change karte hain. Source assertion set me continent–ocean contrast ko winter me greater maana gaya, not summer. [Q90]



EXAM TRAP — Q87–Q89 — Clouds and Heat

- ▶ Clouds outgoing infrared ko simple mirror ki tarah “reflect” karne se zyada efficiently absorb aur re-emit karti hain; downward re-emission night-time cooling ko reduce karti hai.
- ▶ Low thick clouds ka shortwave reflection cooling dominate kar sakta hai, jabki high thin clouds ka longwave greenhouse effect comparatively stronger ho sakta hai.

9. Cold Storage, Wool, Sound in Humid Air & Open Refrigerator

- Cold chamber me low temperature fruit respiration/metabolic rate ko slow karta hai, isliye ripening/spoilage delay aur storage life increase hoti hai. [Q91]
- Woollen clothes mainly fibres ke beech trapped air ki insulating layer ki wajah se warm rakhte hain; wool protein hona primary explanation nahi hai. [Q92]
- Humidity increase hone par air ki effective density decrease hone se sound velocity generally increase hoti hai; therefore “humidity badhne par sound speed decreases” statement incorrect hai. [Q93]
- Closed room me refrigerator ka door open chhodne par room cool nahi hota; compressor electrical work add karta hai aur condenser room me extracted heat + work reject karta hai, so room temperature slightly increase hoti hai. [Q94–Q95]

✓ Source coverage: Q1–Q95